



Friction – Static and Kinetic Friction

When two surfaces in contact are in relative motion a force acts between them parallel to the surface and opposite to the direction of relative motion. This force is known as **frictional force**.

We have **two types of friction**:

- (1) Solid friction (External Friction)
- (2) Liquid friction (Internal friction /viscosity)

Solid friction can be divided into two

- (1) Static friction
- (2) Kinetic friction

Kinetic friction friction is again divided into two

- (1) Sliding friction
- (2) Rolling friction

Static Friction: A block is kept on the table a small force P is applied on the block. The block does not move. This indicates that the resultant force on the block is zero i.e. the applied force P must be balanced by frictional force f .

If the applied force is increased slightly the block still remain at rest which suggest that the frictional force has also equally increases. This frictional force is known as static frictional force.

Thus when no force is applied on the block static frictional force is zero. The moment an external force (small) is applied on the block static frictional force also comes into play balancing the applied force. In this way static frictional force reaches to the maximum value beyond which it can't increase anymore. When the applied force becomes equal or greater than the maximum value of static frictional force the block starts moving.

Thus static frictional force is not a single valued force. It is multi valued force having a continuous range from 0 to f_s maximum, where f_s stand for the static friction. Generally we represent the static frictional force by f_s only which means the maximum value.

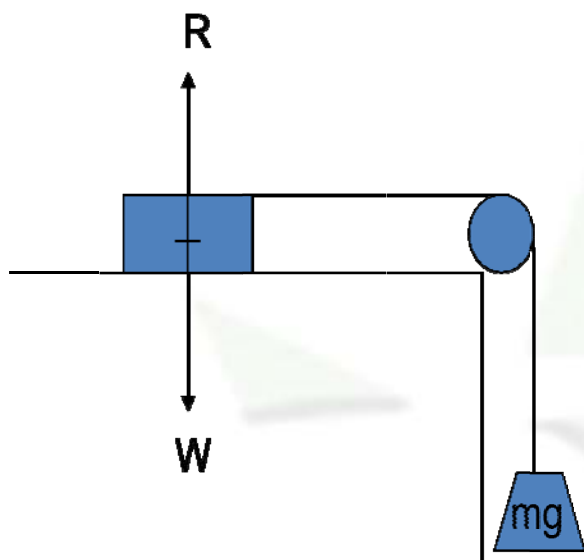
Once the body is set into motion the frictional force between the two surfaces slightly decreases and this force is known as **kinetic friction** ($f_s > f_k$)



Friction – Static and Kinetic Friction

Coefficient of static friction:

A block is kept on the table a cord is tied with over a pulley and carries a hanger at its free end.



Given: W = Weight of the block

Let R = Normal reaction of the table on the block

Since block is in vertical equilibrium

$$W = R$$

Load is put on the hanger and is gradually increases till the block just starts to move (i.e. in such a position where the block is just to slide).

If f_s = Static frictional force (maximum value) between the block and the table then $mg = f_s$, Putting more load on the block and increasing the on the hanger

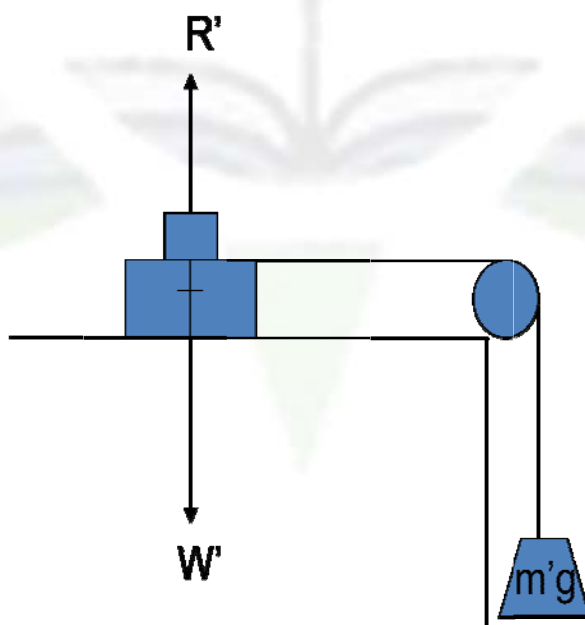
$$W' = R'$$

$$m'g = f_s'$$

$$\frac{W'}{W} = \frac{R'}{R}$$

$$\frac{R'}{R} = \frac{f_s'}{f_s}$$

$$f_s \propto R$$





Friction – Static and Kinetic Friction

From these experimental values it is found that maximum value of static frictional force is proportional to the normal reaction.

$$f_s \propto R$$

$$f_s = \mu_s R$$

$$\mu_s = \frac{f_s}{R}$$

Where μ_s is constant of proportionality and is known as the coefficient of static friction.

Thus coefficient of static friction can be defined as the ratio of the maximum value of static frictional force to the normal reaction.

Similarly coefficient of kinetic friction can also be defined as the ratio of kinetic frictional force to the normal reaction.

$$f_k \propto R$$

$$f_k = \mu_k R$$

$$\mu_k = \frac{f_k}{R}$$